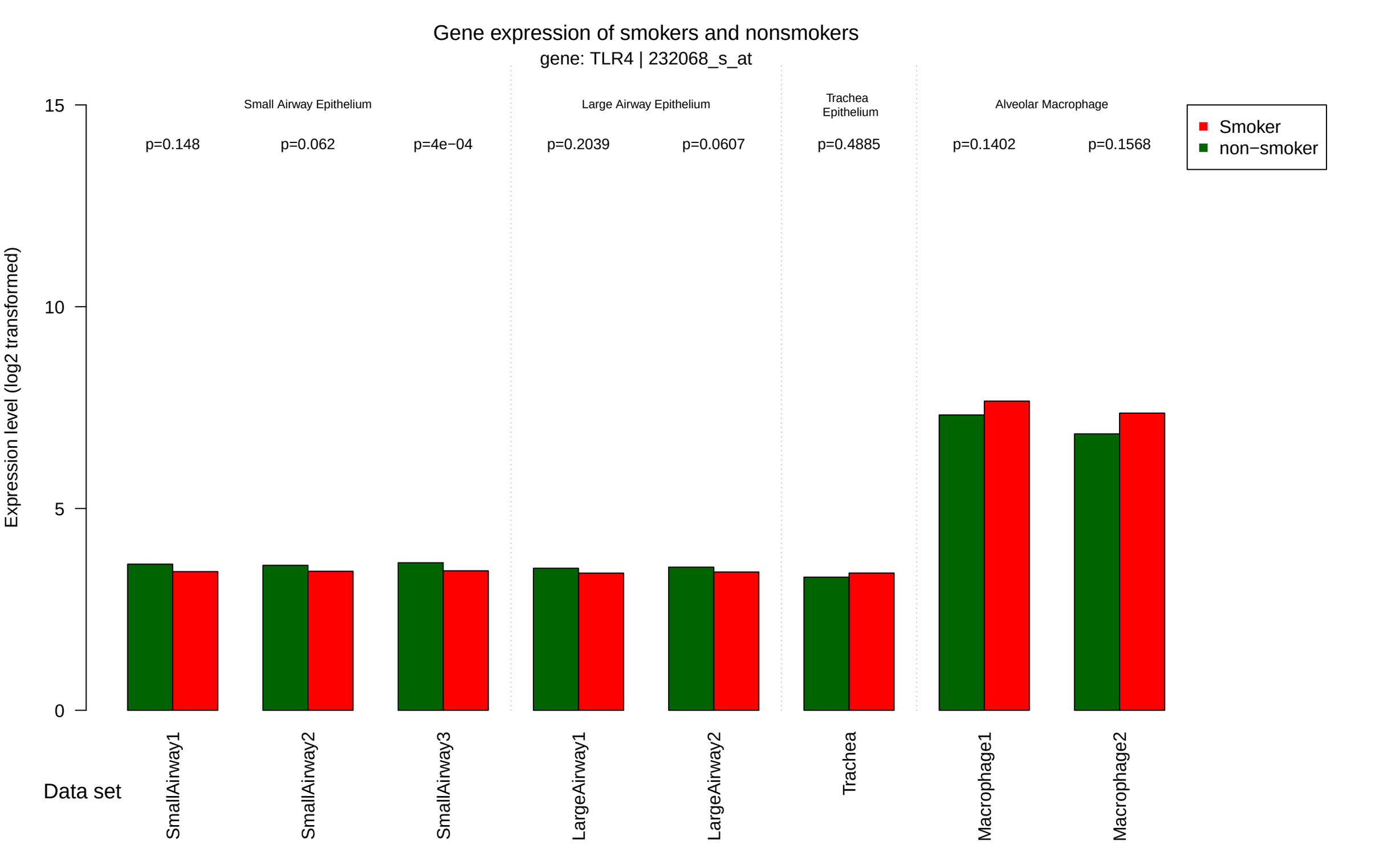
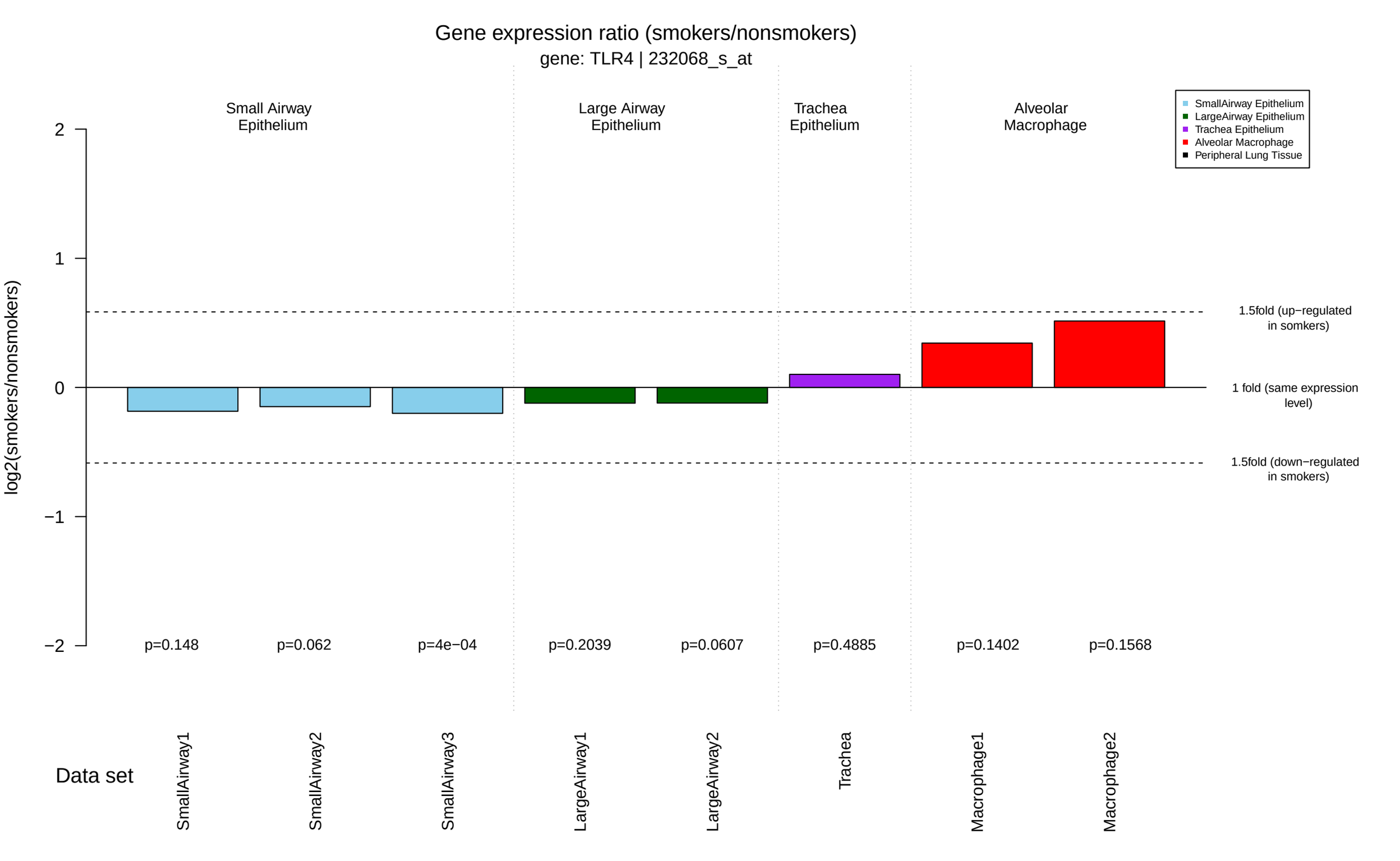
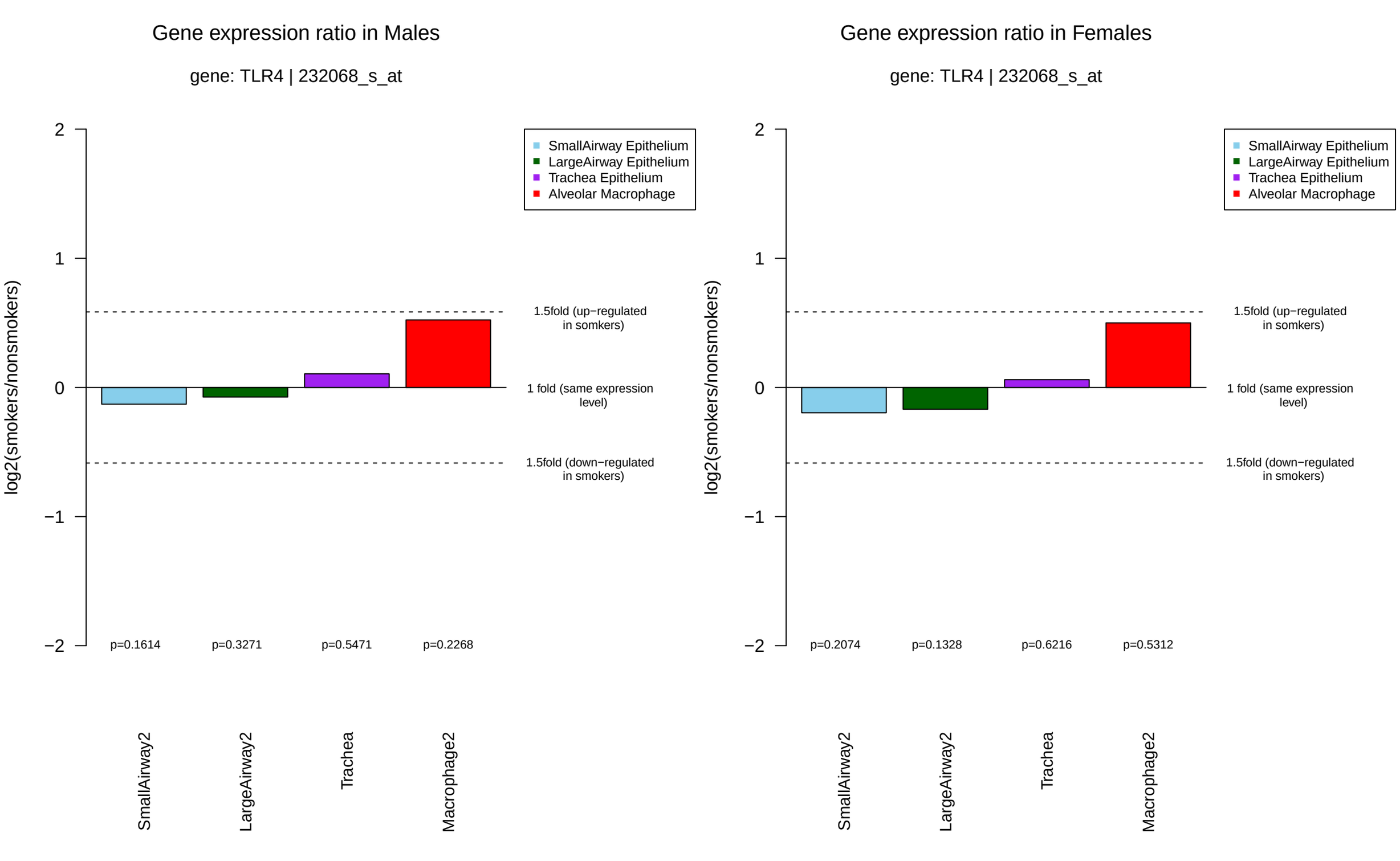
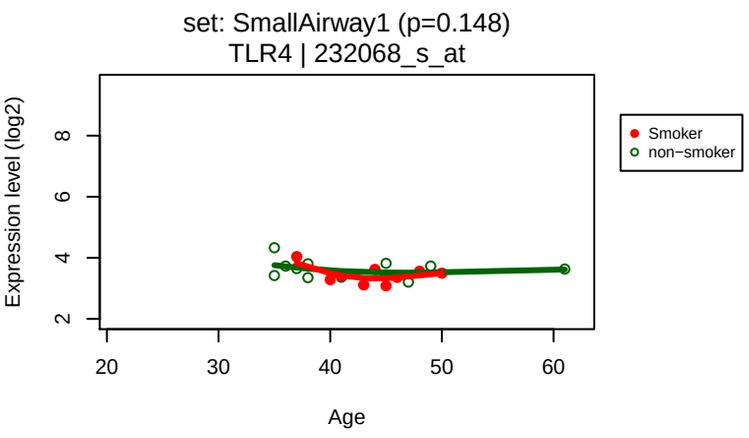
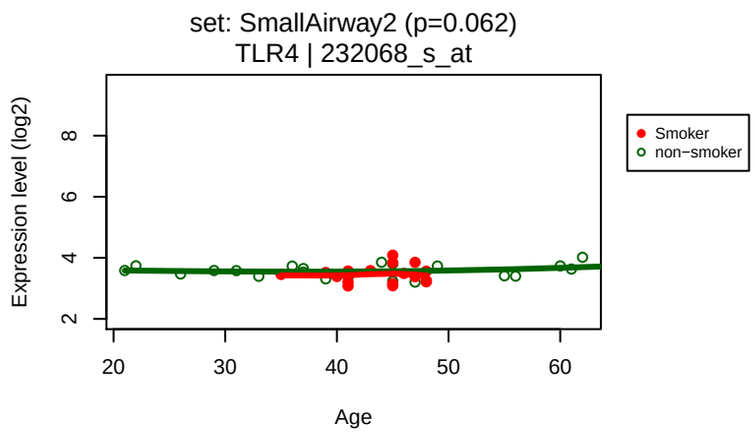
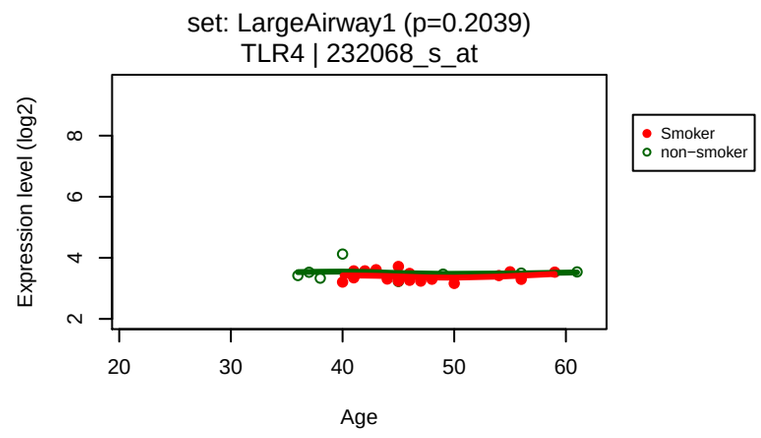

Gene expression of smokers and nonsmokers
gene: TLR4 | 232068_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.148
p=0.062
p=4e−04
p=0.2039
p=0.0607
p=0.4885
p=0.1402
p=0.1568
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

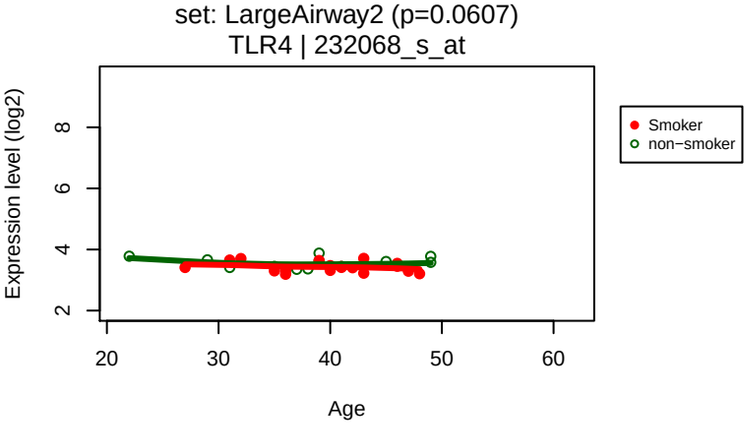

Gene expression ratio (smokers/nonsmokers)
gene: TLR4 | 232068_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.148
p=0.062
p=4e−04
p=0.2039
p=0.0607
p=0.4885
p=0.1402
p=0.1568
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

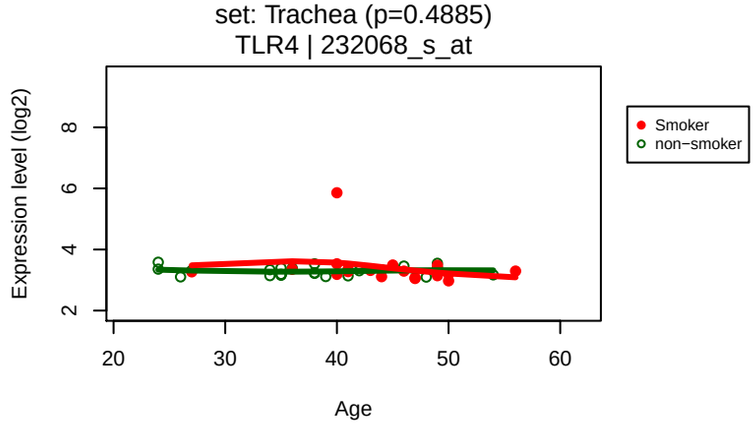

Gene expression ratio in Males
gene: TLR4 | 232068_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1614
p=0.3271
p=0.5471
p=0.2268
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TLR4 | 232068_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2074
p=0.1328
p=0.6216
p=0.5312
SmallAirway2
LargeAirway2
Trachea
Macrophage2

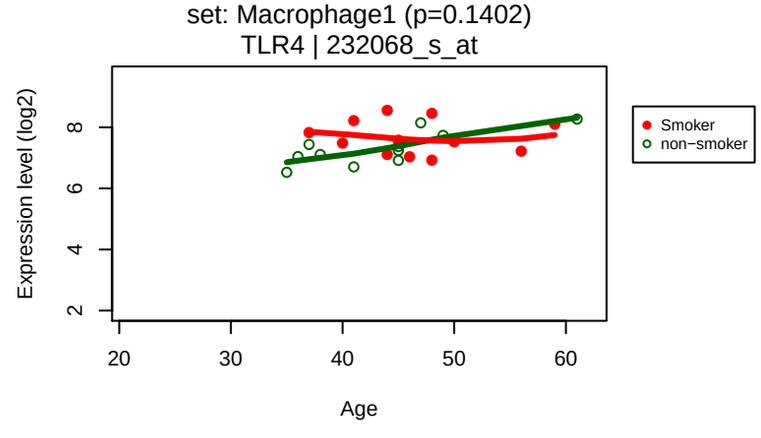

set: SmallAirway1 (p=0.148)
TLR4 | 232068_s_at
Expression level (log2)
Age
Smoker
non−smoker

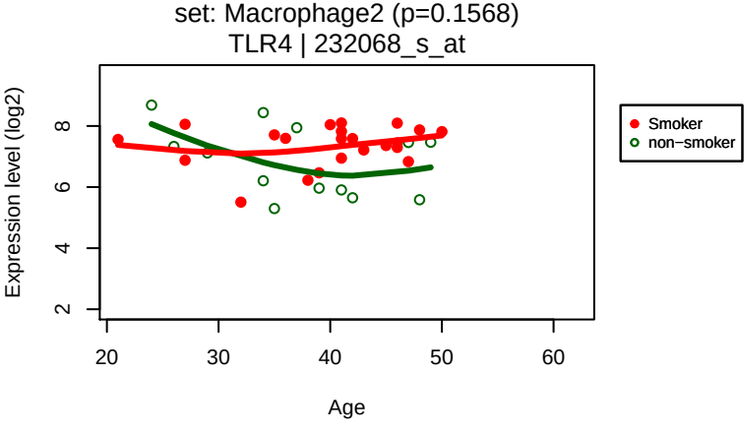

set: SmallAirway2 (p=0.062)
TLR4 | 232068_s_at
Expression level (log2)
Age
Smoker
non−smoker

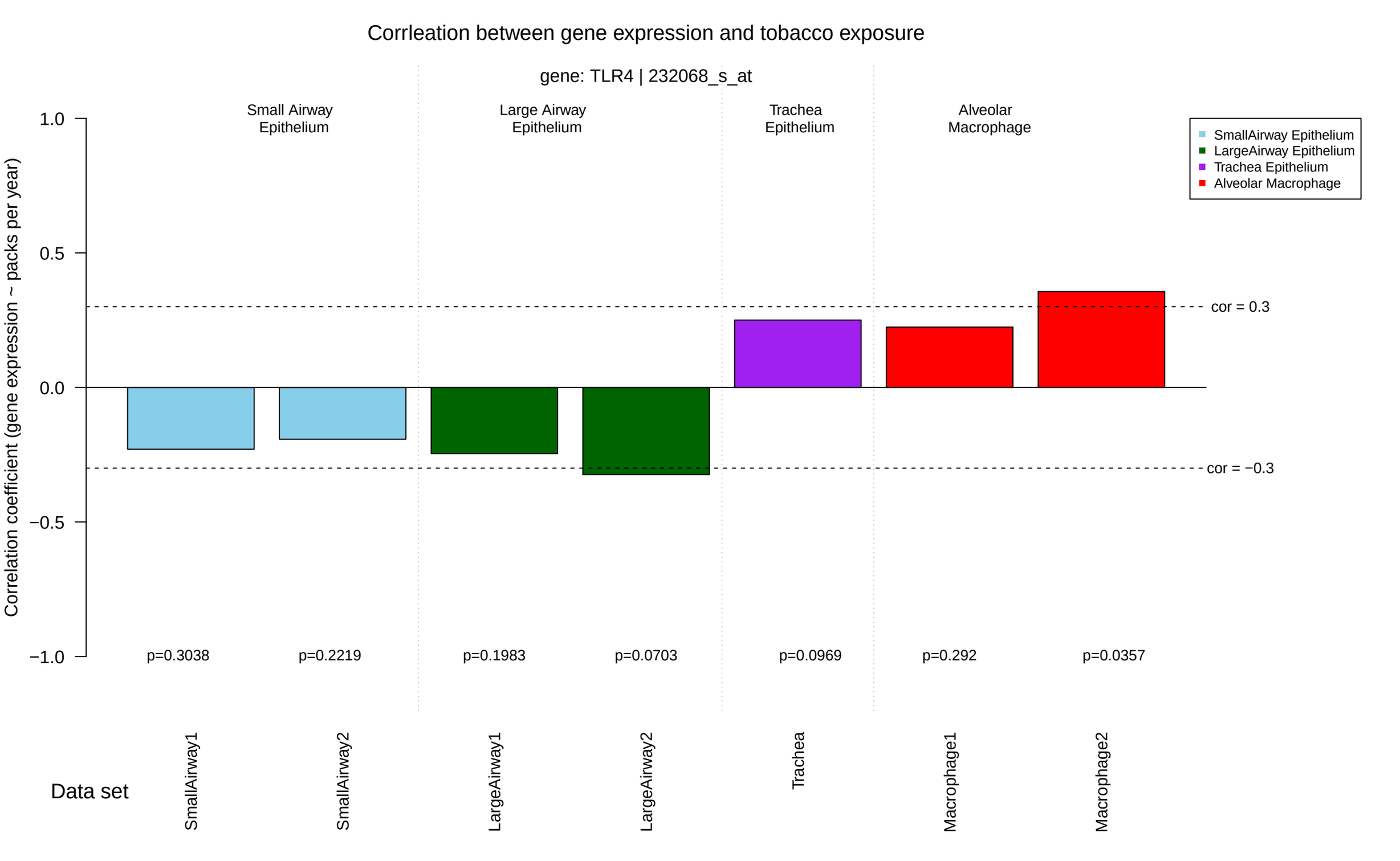

set: LargeAirway1 (p=0.2039)
TLR4 | 232068_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0607)
TLR4 | 232068_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4885)
TLR4 | 232068_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1402)
TLR4 | 232068_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1568)
TLR4 | 232068_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TLR4 | 232068_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3038
p=0.2219
p=0.1983
p=0.0703
p=0.0969
p=0.292
p=0.0357
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2